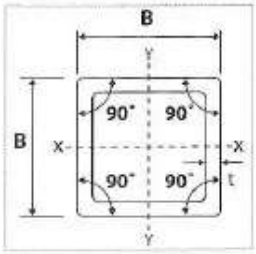
6d (i) Carbon Steel Square Pipe For General Structural Purpose

(Dimension and Properties in accordance to JIS G 3466 : 2066)

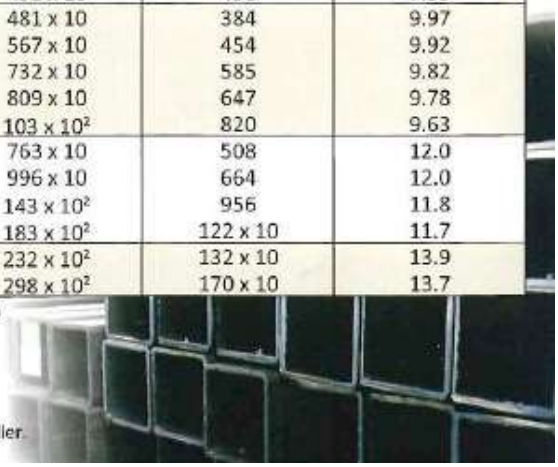


Square Pipe

JIS G 3466

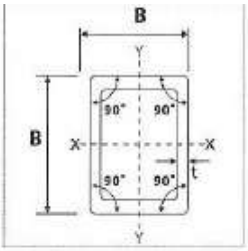
Size	Wall Thickness	Weight	Cross Section	Geometric Moment Of Inertia	Section Modulus	Radius Of Gyration Of Cross-Section
$B \times B$	t	W	A	I_X, I_Y	Z_X, Z_Y	i_X, i_Y
mm	mm	kg / m	cm^2	cm^4	cm^3	cm
40 x 40	1.6	1.88	2.392	5.79	2.90	1.56
	2.3	2.62	3.332	7.73	3.86	1.52
50 x 50	1.6	2.38	3.032	11.7	4.68	1.96
	2.3	3.34	4.252	15.9	6.34	1.93
	3.2	4.50	5.727	20.4	8.16	1.89
60 x 60	1.6	2.88	3.672	20.7	6.89	2.37
	2.3	4.06	5.172	28.3	9.44	2.34
	3.2	5.50	7.007	36.9	12.3	2.30
75 x 75	1.6	3.64	4.632	41.3	11.0	2.99
	2.3	5.14	6.552	57.1	15.2	2.95
	3.2	7.01	8.927	75.5	20.1	2.91
	4.5	9.55	12.17	98.6	26.3	2.85
80 x 80	2.3	5.50	7.012	69.9	17.5	3.16
	3.2	7.51	9.567	92.7	23.2	3.11
	4.5	10.3	13.07	122	30.4	3.05
90 x 90	2.3	6.23	7.932	101	22.4	3.56
	3.2	8.51	10.85	135	29.9	3.52
100 x 100	2.3	6.95	8.852	140	27.9	3.97
	3.2	9.52	12.13	187	37.5	3.93
	4.0	11.7	14.95	226	45.3	3.89
	4.5	13.1	16.67	249	49.9	3.87
	6.0	17.0	21.63	311	62.3	3.79
	9.0	24.1	30.67	408	81.6	3.65
	12.0	30.2	38.53	471	94.3	3.50
125 x 125	3.2	12.0	15.33	376	60.1	4.95
	4.5	16.6	21.17	506	80.9	4.89
	5.0	18.3	23.36	553	88.4	4.86
	6.0	21.7	27.63	641	103	4.82
	9.0	31.1	39.67	865	138	4.67
	12.0	39.7	50.53	103 x 10	165	4.52
150 x 150	4.5	20.1	25.67	896	120	5.91
	5.0	22.3	28.36	982	131	5.89
	6.0	26.4	33.63	115 x 10	153	5.84
	9.0	38.2	48.67	158 x 10	210	5.69
175 x 175	4.5	23.7	30.17	145 x 10	166	6.93
	5.0	26.2	33.36	159 x 10	182	6.91
	6.0	31.1	39.63	186 x 10	213	6.86
200 x 200	4.5	27.2	34.67	219 x 10	219	7.95
	6.0	35.8	45.63	283 x 10	283	7.88
	8.0	46.9	59.79	362 x 10	362	7.78
	9.0	52.3	66.67	399 x 10	399	7.73
	12.0	67.9	86.53	498 x 10	498	7.59
250 x 250	5.0	38.0	48.36	481 x 10	384	9.97
	6.0	45.2	57.63	567 x 10	454	9.92
	8.0	59.5	75.79	732 x 10	585	9.82
	9.0	66.5	84.67	809 x 10	647	9.78
	12.0	86.8	110.5	103 x 10^2	820	9.63
300 x 300	4.5	41.3	52.67	763 x 10	508	12.0
	6.0	54.7	69.63	996 x 10	664	12.0
	9.0	80.6	102.7	143 x 10^2	956	11.8
	12.0	106	134.5	183 x 10^2	122 x 10	11.7
350 x 350	9.0	94.7	120.7	232 x 10^2	132 x 10	13.9
	12.0	124	158.5	298 x 10^2	170 x 10	13.7

The weight values in the table have been obtained by calculating according to the following formula as assuming 1 cm^3 of steel to be 7.85 g
and rounding off the result to three significant figures according to rule A of JIS Z 8401.
$W = 0.015\,7\,t\,(B + B - 3.287\,t)$ where, W : Weight of square tubes (kg / m)
t : Thickness of square tubes (mm) B, B : Size of square tubes (mm)
Other dimensions than given in above table shall be in accordance with the purchaser and the supplier.



6d (ii) Carbon Steel Rectangular Pipe For General Structural Purpose

(Dimension and Properties in accordance to JIS G 3466 : 2066)



Rectangular Pipe

JIS G 3466

Size	Wall Thickness	Weight	Cross Sectional Area	Geometric Moment Of Moment Of Inertia		Section Modulus		Radius Of Gyration Of Cross-Section	
H x B	t	W	A	I_x	I_y	Z_x	Z_y	i_x	i_y
mm	mm	kg / m	cm²	cm⁴	cm⁴	cm³	cm³	cm	cm
50 x 20	1.6	1.63	2.072	6.08	1.42	2.43	1.42	1.71	0.829
	2.3	2.25	2.872	8.00	1.83	3.20	1.83	1.67	0.798
50 x 30	1.6	1.88	2.392	7.96	3.60	3.18	2.40	1.82	1.23
	2.3	2.62	3.332	10.6	4.76	4.25	3.17	1.79	1.20
60 x 30	1.6	2.13	2.712	12.5	4.25	4.16	2.83	2.15	1.25
	2.3	2.98	3.792	16.8	5.65	5.61	3.76	2.11	1.22
	3.2	3.99	5.087	21.4	7.08	7.15	4.72	2.05	1.18
75 x 20	1.6	2.25	2.872	17.6	2.10	4.69	2.10	2.47	0.855
	2.3	3.16	4.022	23.7	2.73	6.31	2.73	2.43	0.824
75 x 45	1.6	2.88	3.672	28.4	12.9	7.56	5.75	2.78	1.88
	2.3	4.06	5.172	38.9	17.6	10.4	7.82	2.74	1.84
	3.2	5.50	7.007	50.8	22.8	13.5	10.1	2.69	1.80
80 x 40	1.6	2.88	3.672	30.7	10.5	7.68	5.26	2.89	1.69
	2.3	4.06	5.172	42.1	14.3	10.5	7.14	2.85	1.66
	3.2	5.50	7.007	54.9	18.4	13.7	9.21	2.80	1.62
90 x 45	2.3	4.60	5.862	61.0	20.8	13.6	9.22	3.23	1.88
	3.2	6.25	7.967	80.2	27.0	17.8	12.0	3.17	1.84
100 x 20	1.6	2.88	3.672	38.1	2.78	7.61	2.78	3.22	0.870
	2.3	4.06	5.172	51.9	3.64	10.4	3.64	3.17	0.839
100 x 40	1.6	3.38	4.312	53.5	12.9	10.7	6.44	3.52	1.73
	2.3	4.78	6.092	73.9	17.5	14.8	8.77	3.48	1.70
	4.2	8.32	10.60	120	27.6	24.0	10.6	3.36	1.61
100 x 50	1.6	3.64	4.632	61.3	21.1	12.3	8.43	3.64	2.13
	2.3	5.14	6.552	84.8	29.0	17.0	11.6	3.60	2.10
	3.2	7.01	8.927	112	38.0	22.5	15.2	3.55	2.06
	4.5	9.55	12.17	147	48.9	29.3	19.5	3.47	2.00
125 x 40	1.6	4.01	5.112	94.4	15.8	15.1	7.91	4.30	1.76
	2.3	5.69	7.242	131	21.6	20.9	10.8	4.25	1.73
125 x 75	2.3	6.95	8.852	192	87.5	30.6	23.3	4.65	3.14
	3.2	9.52	12.13	257	117	41.1	31.1	4.60	3.10
	4.0	11.7	14.95	311	141	49.7	37.5	4.56	3.07
	4.5	13.1	16.67	342	155	54.8	41.2	4.53	3.04
	6.0	17.0	21.63	428	192	68.5	51.1	4.45	2.98
150 x 75	3.2	10.8	13.73	402	137	53.6	36.6	5.41	3.16
150 x 80	4.5	15.2	19.37	563	211	75.0	52.9	5.39	3.30
	5.0	16.8	21.36	614	230	81.9	57.5	5.36	3.28
	6.0	19.8	25.23	710	264	94.7	66.1	5.31	3.24
150 x 100	3.2	12.0	15.33	488	262	65.1	52.5	5.64	4.14
	4.5	16.6	21.17	658	352	87.7	70.4	5.58	4.08
	6.0	21.7	27.63	835	444	111	88.8	5.50	4.01
	9.0	31.1	39.67	113 x 10	595	151	119	5.33	3.87
200 x 100	4.5	20.1	25.67	133 x 10	455	133	90.9	7.20	4.21
	6.0	26.4	33.63	170 x 10	577	170	115	7.12	4.14
	9.0	38.2	48.67	235 x 10	782	235	156	6.94	4.01
200 x 150	4.5	23.7	30.17	176 x 10	113 x 10	176	151	7.64	6.13
	6.0	31.1	39.63	227 x 10	146 x 10	227	194	7.56	6.06
	9.0	45.3	57.67	317 x 10	202 x 10	317	270	7.41	5.93
250 x 150	6.0	35.8	45.63	389 x 10	177 x 10	311	236	9.23	6.23
	9.0	52.3	66.67	548 x 10	247 x 10	438	330	9.06	6.09
	12.0	67.9	86.53	685 x 10	307 x 10	548	409	8.90	5.95
300 x 200	6.0	45.2	57.63	737 x 10	396 x 10	491	396	11.3	8.29
	9.0	66.5	84.67	105 x 10^2	563 x 10	702	563	11.2	8.16
	12.0	86.8	110.5	134 x 10^2	711 x 10	890	711	11.0	8.02
350 x 150	6.0	45.2	57.63	891 x 10	239 x 10	509	319	12.4	6.44
	9.0	66.5	84.67	127 x 10^2	337 x 10	726	449	12.3	6.31
	12.0	86.8	110.5	161 x 10^2	421 x 10	921	562	12.1	6.17
400 x 200	6.0	54.7	69.63	148 x 10^2	509 x 10	739	509	14.6	8.55
	9.0	80.6	102.7	213 x 10^2	727 x 10	107 x 10	727	14.4	8.42
	12.0	106	134.5	273 x 10^2	923 x 10	136 x 10	923	14.2	8.23

The weight values in the table have been obtained by calculating according to the following formula as assuming 1 cm³ of steel to be 7.85 g and rounding off the result to three significant figures according to rule A of JIS Z 8401.

$W = 0.015\,7\,t\,(H + B - 3.287\,t)$ where, W: Weight of rectangular tubes (kg / m) t: Thickness of rectangular tubes (mm) H, B: Size of rectangular tubes (mm)

Other dimensions than given in above table shall be in accordance with the purchaser and the supplier.

6d) Carbon Steel Square And Rectangular Tubes For General Structure

Japanese Industrial Standard (Extracts from JIS G 3466 : 2006)

General Information

This Japanese Industrial Standard specifies the carbon steel square and rectangular tubes (hereafter referred to as "square and rectangular tubes") used for civil engineering, architecture and other structures. Square and rectangular tubes are classified into two classes and the symbols of the classes shall be as shown in Table 1.

Table 1. Class symbols

Class Symbol
STKR 400
STKR 490

Chemical Composition

The Heat analysis values and Product analysis values shall satisfy the requirements given in Table 2.

Table 2. Chemical Composition

Unit: %

Class Symbol	C	Si	Mn	P	S
STKR 400	0.25 max.	-	-	0.040 max.	0.040 max.
STKR 490	0.18 max.	0.55 max.	1.50 max.	0.040 max.	0.040 max.

Mechanical Strength (Tensile Test)

The resulting values of tensile strength, yield point or proof stress and elongation shall satisfy the requirements in Table 3. However, for square and rectangular tubes under 8 mm in wall thickness, the minimum elongation as a result of the tensile test using *No.5 test piece shall be in according with Table 4.

Table 3. Mechanical properties

Unit: %

Class Symbol	Tensile Strength N/mm^2	Yield Point Or Proof Stress N/mm^2	Elongation %
			*No. 5 Test Piece
STKR 400	400 min.	245 min.	23 min.
STKR 490	490 min.	325 min.	23 min.

NOTE: 1 N/mm^2 = 1 MPa

Table 4. Minimum elongation for square and rectangular tubes of thickness under 8 mm as a result of test using No.5 test piece

Unit: %

Class Symbol	Test Piece	Division Of Wall Thickness							
		1 mm max.	Over 1mm up to and incl. 2 mm	Over 2mm up to and incl. 3 mm	Over 3mm up to and incl. 4 mm	Over 4mm up to and incl. 5 mm	Over 5mm up to and incl. 6 mm	Over 6mm up to and incl. 7 mm	Over 7mm to and excl. 8 mm
STKR 400	*No. 5 Test Piece	12	14	16	17	18	20	22	23
STKR 490									

NOTE: The above value have been obtained by subtracting 1.5 from the values in table 3 per 1 mm decreament from the 8 mm tube wall thickness, and then rounding the results off to an integer in accordance with rule A of JIS Z 8401

* Please refer to Appendix 'A' Tension Test Pieces for Metallic Materials - JIS Z 2201 (pg153-155)

Tolerances On Dimensions and Mass

The tolerances on dimensions shall be as specified in Table 5.

Table 5. Tolerance on dimensions

Specified Item And Dimension			Dimensional Tolerances
Side length	100 mm or under		± 1.5 mm
	Over 100 mm		± 1.5 %
Unevenness of the flat portion of each side	Side length 100 mm or under		0.5 mm or under
	Side length over 100 mm		0.5 % or less of the side length
Thickness	Square and rectangular tubes manufactured by welding	Under 3 mm	± 0.3 mm
		3 mm or over	± 10 %
	Seamless square and rectangular tubes	Under 4 mm	± 0.6 mm
		4 mm or over	± 15 %
Length			+ not specified 0
Straightness			0.3 % or less of the whole length
Out-of-roundness			-
Concavity / convexity			-
Radius of Corners			3 *t* or under
Squareness of side			-
Twist			-
Angularity between the adjacent flat portions			± 1.5°
Inner Flash			-
End tolerance on diameter			-
End Facing			-
Pipe End			-
Mass (*m*) per unit length			-

